

Test Case: Take Onboarding Survey

TC-AISPEAKING-001

Title: Open AI Speaking page with default scenario

Preconditions:

- User is logged in and has completed onboarding
- Backend is running

Steps:

1. Open the **AI Speaking** page from the sidebar
2. Check the scenario list, question area, and conversation history

Expected Result:

- Page loads with three scenarios: **Technical English**, **Daily Tech Sync**, and **Tech Interviews**
- **Technical English** is selected by default
- A speaking question is shown
- The question also appears in Conversation History
- Score cards show no score yet
- Error detection and feedback is empty
- **Start Listening** is enabled

Priority: High

TC-AISPEAKING-002

Title: Change scenario to Daily Tech Sync

Preconditions:

- User is on the AI Speaking page
- A question is already loaded

Steps:

1. Click **Daily Tech Sync**
2. Check the new question and conversation history

Expected Result:

- A new daily-sync question is shown
- Conversation History is reset with the new question
- Previous answer, scores, and feedback are cleared
- **Start Listening** is enabled

Priority: High

TC-AISPEAKING-003

Title: Change scenario to Tech Interviews

Preconditions:

- User is on the AI Speaking page

Steps:

1. Click **Tech Interviews**
2. Check the new question and conversation history

Expected Result:

- A new interview question is shown
- Conversation History shows the new question
- Score cards show no score yet
- Error detection and feedback is empty

Priority: High

TC-AISPEAKING-004

Title: Record answer and see live transcript

Preconditions:

- User is on the AI Speaking page
- A question is loaded
- Microphone access is allowed

Steps:

1. Click **Start Listening**
2. Speak a short answer in English
3. Check the recording state and conversation history

Expected Result:

- **Stop Practice** becomes enabled
- Page shows that recording is active
- Spoken words appear in Conversation History while speaking

Priority: High

TC-AISPEAKING-005

Title: Analyze answer and show feedback

Preconditions:

- User has recorded an answer
- Backend is running

Steps:

1. Click **Stop Practice**
2. Wait for analysis to finish
3. Check the score cards and feedback sections

Expected Result:

- Grammar, Tech Vocabulary, and Clarity scores are shown
- Grammar feedback and improvement tips are shown
- Vocabulary and clarity feedback are shown
- A new question is loaded for the next turn
- The previous question and answer stay in Conversation History

Priority: High

TC-AISPEAKING-006

Title: Stop practice with no speech

Preconditions:

- User is on the AI Speaking page
- A question is loaded
- Microphone access is allowed

Steps:

1. Click **Start Listening**
2. Stay silent
3. Click **Stop Practice**

Expected Result:

- Error message is shown: "**No speech was detected. Please try again.**"
- Scores do not change
- The same question remains for retry

Priority: Medium

TC-AISPEAKING-007

Title: Show grammar errors after analysis

Preconditions:

- User is on the AI Speaking page
- A question is loaded

- Backend is running

Steps:

1. Click **Start Listening**
2. Speak a sentence with a grammar mistake, for example: "I am responsible for manage the database."
3. Click **Stop Practice**
4. Check the grammar feedback section

Expected Result:

- Grammar score is below 100
- The wrong phrase is shown with a correction suggestion
- Explanation and a better example sentence are shown

Priority: High

TC-AISPEAKING-008

Title: Keep conversation history after multiple turns

Preconditions:

- User has completed at least one practice turn

Steps:

1. Complete one full practice turn
2. Complete a second practice turn
3. Check Conversation History

Expected Result:

- Previous questions and answers remain visible
- The latest question is shown for the current turn
- History includes all completed turns

Priority: Medium

TC-AISPEAKING-009

Title: Disable scenario change during analysis

Preconditions:

- User is on the AI Speaking page
- A question is loaded

Steps:

1. Record an answer and click **Stop Practice**

2. While analysis is running, try to click another scenario

Expected Result:

- Scenario buttons cannot be clicked during analysis
- **Start Listening** and **Stop Practice** are disabled during analysis
- Scenario buttons work again after analysis finishes

Priority: Medium

TC-AISPEAKING-010

Title: Show error when backend is unavailable

Preconditions:

- User is logged in
- Backend is stopped or unreachable

Steps:

1. Open the AI Speaking page or switch scenario while backend is down
2. Check the page response

Expected Result:

- Error message is shown: "**Unable to load question. Please check backend connection.**"
- No question is loaded
- **Start Listening** stays disabled

Priority: Medium

Test Execution

Test Case ID	Execution Date	Status	Actual Result	Bug ID
TC-AISPEAKING-001	20/08/2026	Pass	Page loaded with default Technical English scenario and a question	N/A
TC-AISPEAKING-002	20/08/2026	Pass	Switched to Daily Tech Sync and a question is loaded	N/A
TC-AISPEAKING-003	20/08/2026	Pass	Switched to Tech Interviews and a question is loaded	N/A
TC-AISPEAKING-004	21/08/2026	Pass	User's speech is successfully displayed onto Conversation History	N/A
TC-AISPEAKING-005	21/08/2026	Pass	Feedbacks and score are displayed correctly	N/A

Test Case ID	Execution Date	Status	Actual Result	Bug ID
TC-AISPEAKING-006	20/08/2026	Pass	Error message is successfully displayed and score stayed the same	N/A
TC-AISPEAKING-007	21/08/2026	Pass	Mistake is spotted and feedbacks are loaded	N/A
TC-AISPEAKING-008	21/08/2026	Pass	Conversation History properly stored all previous questions and answers	N/A
TC-AISPEAKING-009	21/08/2026	Pass	Scenario menu is successfully blocked	N/A
TC-AISPEAKING-010	20/08/2026	Pass	Error message is successfully displayed	N/A

Notes from Test Run (20/08/2026)

- Random question API works for all three scenarios
- Invalid scenario and empty transcript return error responses
- No bugs found in the tested API paths during this session
- Speech-to-text is working as intended
- Analyze API returns scores and feedback correctly